



Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

BIOLOGY

Paper 2 Multiple Choice (Extended)

0610/21

May/June 2018

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

This syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **15** printed pages and **1** blank page.



- 1 Which organisms carry out respiration, growth, movement and excretion?
- A all animals and all plants
 - B animals only
 - C arthropods and flowering plants only
 - D plants only
- 2 The diagram shows an animal whose scientific name is *Falco peregrinus*.



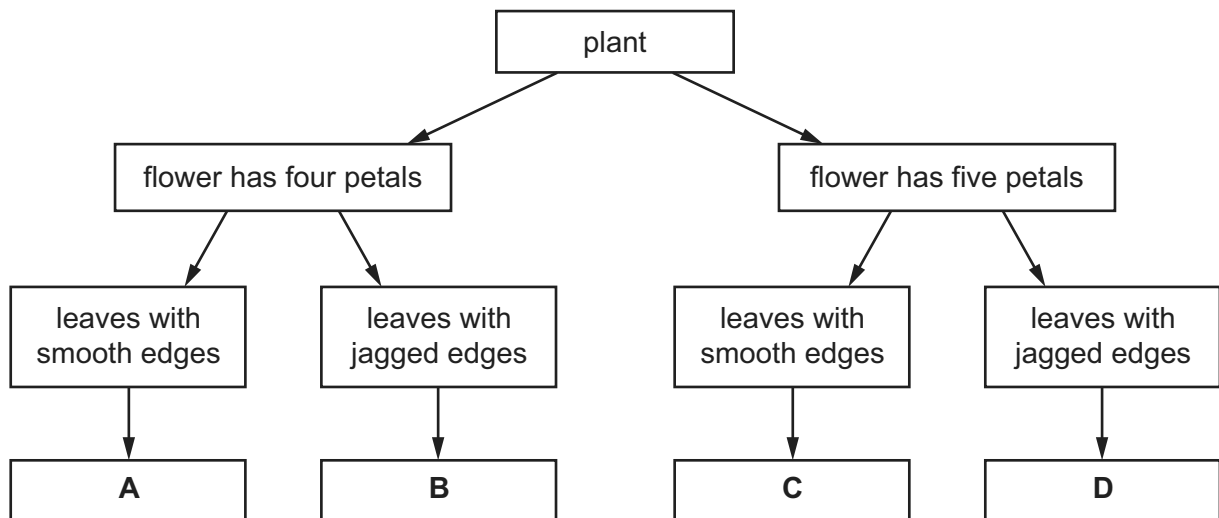
To which species does it belong?

- A bird
 - B *F. peregrinus*
 - C *Falco*
 - D vertebrate
- 3 What kind of skin do amphibians have?
- A dry without scales
 - B dry with scales
 - C moist without scales
 - D moist with scales

- 4 The diagram shows a flowering plant.



Use the key to identify the plant.



- 5 In which part of the cell does aerobic respiration occur?

- A cytoplasm
- B mitochondrion
- C ribosome
- D vesicle

- 6 Why do some root cells have root hairs?

- A for the maintenance of the temperature of the cell sap
- B to increase the surface area of the cells
- C to increase the volume of the cell sap
- D to provide a place for cell nuclei

7 Which words correctly complete the paragraph?

Diffusion may be defined as the net movement of particles from a region of their1..... concentration, to a region of their2..... concentration, where movement is3..... a concentration gradient.

	1	2	3
A	higher	lower	down
B	higher	lower	up
C	lower	higher	down
D	lower	higher	up

8 Which part of a plant root hair is partially permeable?

- A** the cell sap
- B** the cell surface membrane
- C** the cell vacuole
- D** the cell wall

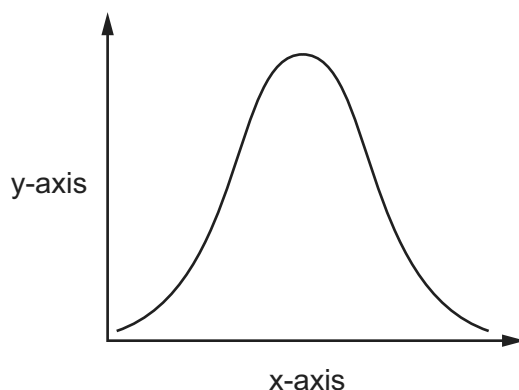
9 The table shows the results of food tests carried out on a fruit.

test	Benedict's	biuret	ethanol	iodine
result	positive	positive	negative	negative

What did the fruit contain?

- A** fat and reducing sugar
- B** fat and starch
- C** protein and reducing sugar
- D** protein and starch

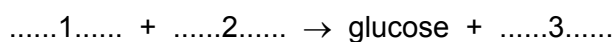
- 10 An experiment was carried out to investigate the effect of pH on enzyme action. The graph shows the results.



What are the labels for the x-axis and the y-axis?

	x-axis	y-axis
A	pH	rate of reaction
B	pH	time
C	rate of reaction	pH
D	time	pH

- 11 The equation for photosynthesis is shown.



Which words correctly complete gaps 1, 2 and 3?

	1	2	3
A	carbon dioxide	light	oxygen
B	carbon dioxide	water	oxygen
C	oxygen	light	carbon dioxide
D	oxygen	water	carbon dioxide

- 12 What is the best source of vitamin C in a balanced diet?

- A** fish
- B** fruit
- C** meat
- D** rice

13 A young, active woman requires more of which constituent in her diet than a young, active man?

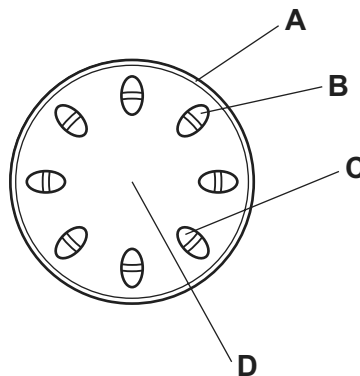
- A** fat
- B** iron
- C** protein
- D** vitamin C

14 Which condition could be caused by a lack of iron?

- A** anaemia
- B** cholera
- C** scurvy
- D** diabetes

15 The diagram shows part of a section through a plant stem.

Which tissue transports water from the roots to the leaves?



16 Petroleum jelly is waterproof and transparent.

Covering the underside of the leaves of a plant with a thin layer of petroleum jelly will slow down the rate of water loss from the plant.

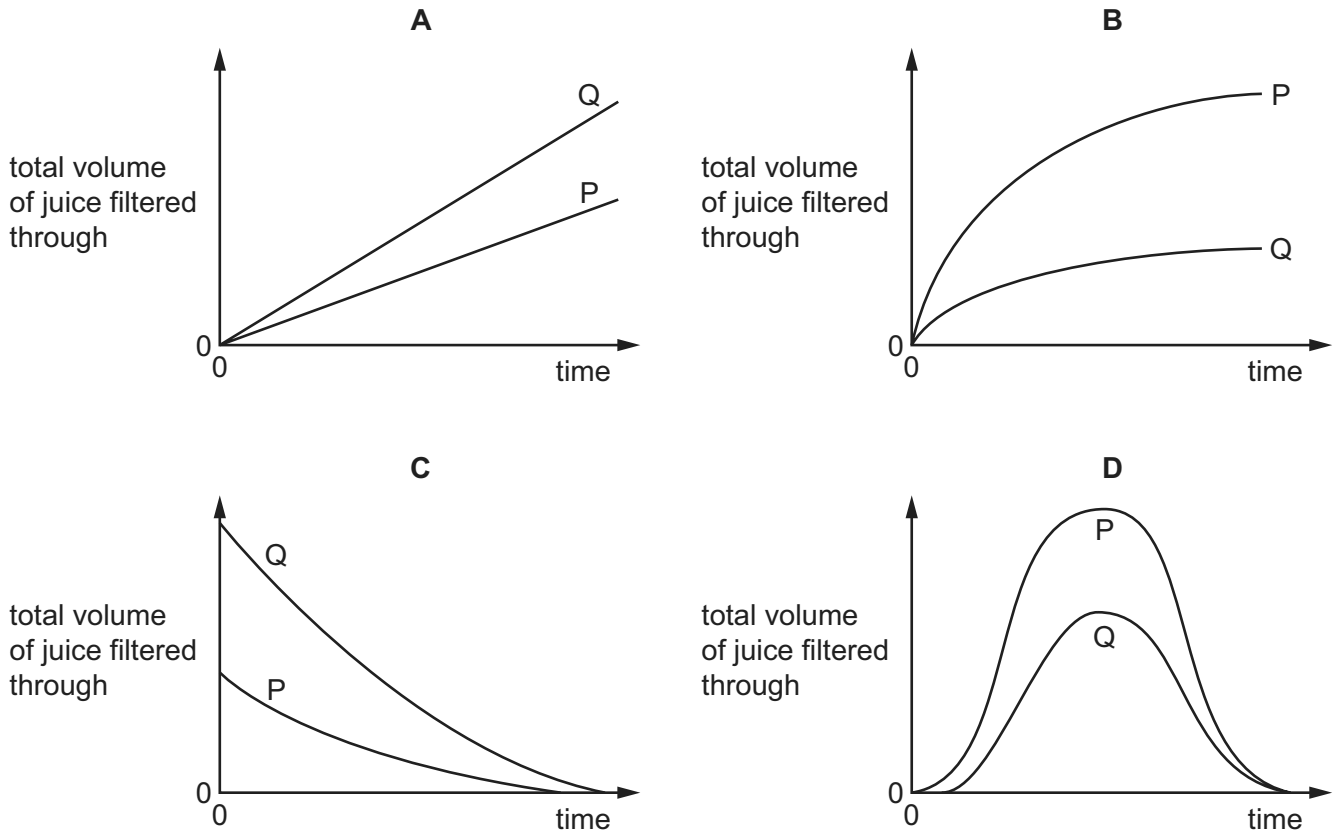
Which statement explains this?

- A** Plants absorb nutrients from the petroleum jelly.
- B** Plants absorb water from the petroleum jelly.
- C** Stomata are blocked by the petroleum jelly.
- D** The petroleum jelly stops photosynthesis.

- 17** A student places two samples of crushed apple into two beakers, P and Q. The samples are of equal size. She adds 5 cm^3 of pectinase solution to beaker P and 5 cm^3 of water to beaker Q.

After five minutes, she places the samples of crushed apple into two different filter funnels, and measures the volume of juice filtering through from each sample over a period of 10 minutes.

Which graph shows her results?



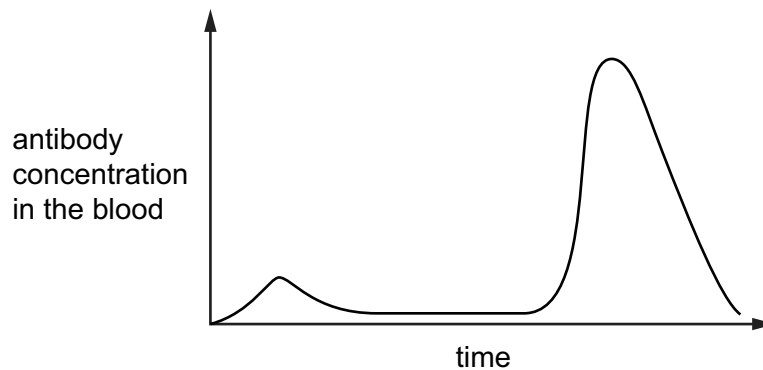
- 18** During the process of blood clotting, damage to blood vessels stimulates L, and M is converted to N.

What are L, M and N?

	L	M	N
A	fibrin	platelets	fibrinogen
B	fibrinogen	platelets	fibrin
C	platelets	fibrin	fibrinogen
D	platelets	fibrinogen	fibrin

- 19** A child is vaccinated against measles. After a period of time the child is infected with the measles virus.

The graph shows the concentration of measles antibodies in the child's bloodstream during this time.



Which statement is consistent with the information in the graph?

- A** After the vaccination, the child produced memory cells.
B The child had passive immunity against measles.
C The measles virus contains antibodies.
D The vaccination failed to protect the child against measles.
- 20** Muscles are responsible for the ventilation of the lungs during breathing.

Which row describes their action during the inspiration of air?

	diaphragm muscles	external intercostal muscles	internal intercostal muscles
A	contract	contract	relax
B	contract	relax	contract
C	relax	contract	relax
D	relax	relax	contract

- 21** Yeast is able to respire both aerobically and anaerobically.

Which statement describes the waste products of yeast respiration?

- A** Aerobic respiration produces alcohol as one of its waste products.
B Aerobic respiration produces three times as much carbon dioxide as anaerobic respiration from one molecule of glucose.
C Anaerobic respiration and aerobic respiration both produce the same amount of carbon dioxide from one molecule of glucose.
D Anaerobic respiration produces three times as much carbon dioxide as aerobic respiration from one molecule of glucose.

- 22** The table shows the presence or absence of chemicals in solution in different parts of a healthy kidney.

Which row is correct?

	chemical	blood plasma in glomerulus	fluid entering kidney tubule	fluid in ureter
A	glucose	✓	x	x
B	protein	✓	✓	✓
C	salts	✓	x	x
D	urea	✓	✓	✓

key

✓ = present

x = absent

- 23** Four processes occur when impulses cross a synapse.

P neurotransmitter diffuses across the gap

Q neurotransmitter binds with receptors

R impulse stimulates vesicles

S release of neurotransmitter

What is the correct sequence for these processes?

A P → R → Q → S

B R → P → Q → S

C R → S → P → Q

D S → Q → R → P

- 24** Which hormone is involved in the conversion of glucose to glycogen?

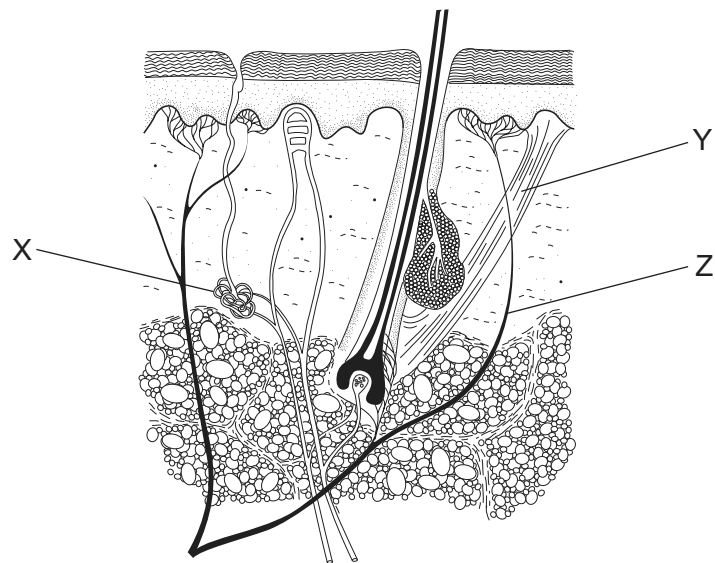
A adrenaline

B insulin

C oestrogen

D testosterone

25 The diagram shows the structure of human skin.

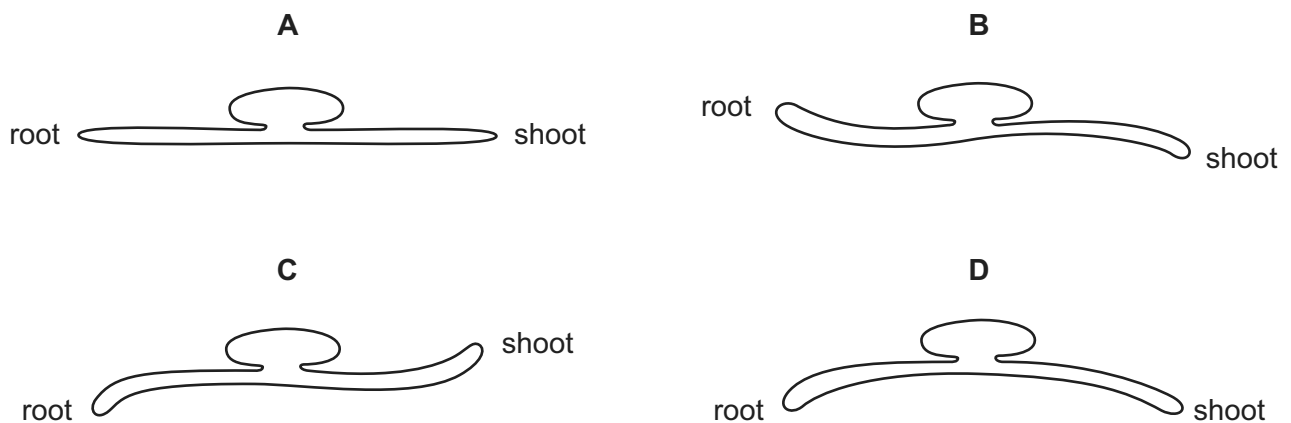


What are X, Y and Z?

	X	Y	Z
A	receptor	sensory neurone	sweat gland
B	sensory neurone	hair erector muscle	receptor
C	sweat gland	hair erector muscle	sensory neurone
D	sweat gland	receptor	blood vessel

26 A seedling was placed in a horizontal position.

Which diagram shows the result of the gravitropic responses in the seedling?



27 Which hormone may be used to improve sporting performance?

- A FSH
- B LH
- C oestrogen
- D testosterone

28 Specific grape varieties are maintained using stem cuttings from mature plants that are then planted and cultivated to produce grapes. This is an example of artificial asexual reproduction.

What is a disadvantage of using asexual reproduction to produce fruit?

- A An outbreak of disease will affect the whole crop in the same way.
- B Genetically identical fruit is produced relatively quickly.
- C No pollination or pollinators are required.
- D The characteristics of the grapes will vary between plants.

29 Which two statements are correct for the process of cross-pollination in plants?

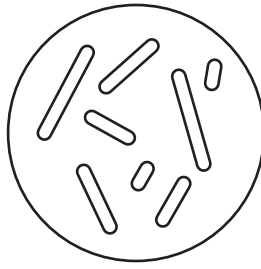
- 1 increases potential for variation in offspring
- 2 pollen is transferred to a different flower on the same plant
- 3 reduces potential to respond to environmental change
- 4 pollen is transferred to a flower on a different plant of the same species

- A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

30 What is a function of the mitochondria in a sperm cell?

- A penetrating the surface of the egg cell
- B propelling the sperm towards the egg
- C storing food energy
- D supplying the energy for movement

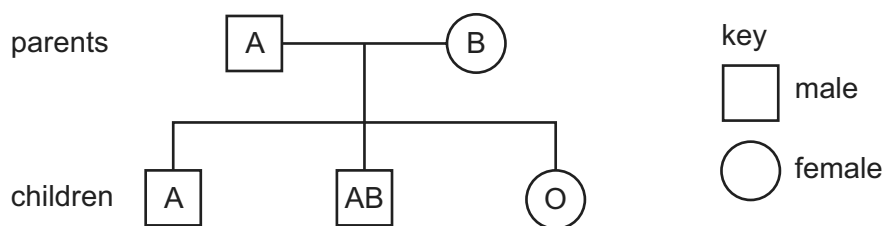
- 31** The diagram shows the chromosomes in the nucleus of a body cell in an adult fruit fly.



What are the diploid and haploid number of chromosomes in the fruit fly?

	diploid	haploid
A	4	8
B	8	4
C	8	16
D	16	8

- 32** The diagram shows the phenotypes for blood groups in a family.



Which statement about the genotypes of the parents is correct?

- A** Both parents have alleles for blood group A and B.
- B** Both parents have the allele for blood group O.
- C** Only the father has the allele for blood group O.
- D** Only the mother has the allele for blood group O.
- 33** Sickle-cell anaemia is a genetic disorder which results in severe illness in homozygous individuals. In some human populations being heterozygous can be beneficial.

What could be the reason for this?

- A** Heterozygous individuals are not affected by the disorder.
- B** Heterozygous individuals are more resistant to malaria.
- C** It is caused by a dominant allele.
- D** The disorder is sex-linked.

34 Which statement describes the relationship between evolution and natural selection?

- A A change in the adaptive features of a population over time causes evolution, resulting in natural selection.
- B Evolution causes a change in the adaptive features of a population over time, resulting in natural selection.
- C Evolution causes natural selection, resulting in a change in the adaptive features of a population over time.
- D Evolution is the change in the adaptive features of a population over time as a result of natural selection.

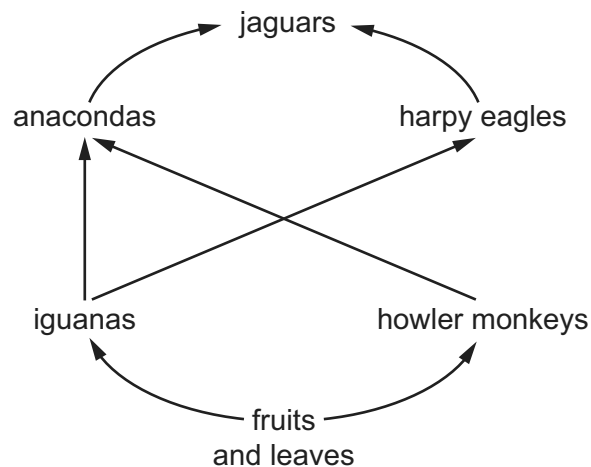
35 The diagram shows a food chain.

maize	→	locusts	→	lizards	→	snakes
10 000 kJ		1000 kJ		100 kJ		10 kJ

What is the efficiency of energy transfer between the maize and the lizards in this food chain?

- A 0.01% B 0.1% C 1% D 10%

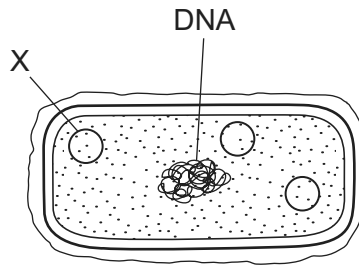
36 The diagram shows part of a food web from a rainforest.



In this food web, at which trophic level are the anacondas?

- A primary consumers
- B secondary consumers
- C tertiary consumers
- D quaternary consumers

37 The diagram shows the structure of a bacterial cell.



The presence of structure X in the bacterial cell is one reason why bacteria are used in genetic engineering.

What is structure X?

- A endoplasmic reticulum
- B mitochondria
- C plasmid
- D ribosome

38 What does *Penicillium* need to grow in a fermenter?

	amino acids	carbohydrates	oxygen
A	✓	✓	✓
B	✓	✓	x
C	✓	x	✓
D	x	✓	✓

39 Deforestation can have a negative impact on the environment.

Which statement about the negative impact of deforestation is correct?

- A decreases the levels of carbon dioxide which can lead to reduced rates of photosynthesis
- B decreases the amount of water flowing in local rivers so there is less flooding
- C leads to soil loss so there is less fertile soil for the growth of crops in the area
- D provides less land for the extraction of natural resources

40 When nitrates enter a lake they cause rapid growth of algae on the surface of the water. This causes the following changes in the lake:

- 1 a decrease in the concentration of dissolved oxygen in the water
- 2 fish and other aquatic animals die
- 3 an increase in aerobic respiration by decomposers
- 4 producers die and decomposition increases

In which order do these changes occur?

- A** 1 → 2 → 4 → 3
- B** 3 → 1 → 2 → 4
- C** 4 → 2 → 3 → 1
- D** 4 → 3 → 1 → 2

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.